Discrimination and Deterrence with Enforcer Liability

Murat C. Mungan,
Antonin Scalia Law School,
George Mason University

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Discrimination and Deterrence with Enforcer Liability

Murat C. Mungan*

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Abstract

Taste-based discrimination (i.e. discrimination due to racist preferences) receives more attention than statistical discrimination in the enforcement literature, because the latter allows enforcers to increase their 'success rates'. I show here that when enforcers' incentives can be altered via liabilities and rewards, all types of discrimination reduce deterrence. Moreover, adverse effects of statistical discrimination on deterrence are more persistent than similar effects due to taste-based discrimination. I identify crime minimizing liabilities and rewards when enforcers engage in racial discrimination, and consider the robustness of the analysis in alternative settings.

Keywords: Racial profiling, statistical discrimination, taste-based discrimination, law enforcer incentives, crime, deterrence.

JEL classification: K00, K14, K42, J14, J15.

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I don't think the kind of boy he is has anything to do with it. The facts are supposed to determine the case.

-Juror #9,
12 Angry Men

1 Introduction

Several high profile cases of black men being killed or shot by law enforcers has once again made racial discrimination a frequently debated topic. In the course of these debates, many people refer to statistics suggesting that African Americans are more frequently stopped and/or searched by law enforcers.[1] Some people claim that the large differences between these rates indicate racial discrimination. Others, including some police departments, point out that these differences can be the result of large differences between the crime rates among the black and white populations. A large body of academic work has also informed this debate.

The academic literature points out that, differences in search rates do not necessarily imply racial discrimination in law enforcement. This is because, if members of one race more frequently engage in criminal activity than members of another race, and, as long as guilty people, on average, appear more suspicious, law enforcers will be inclined to search members of the former race more often, simply because a greater percentage of them appear suspicious. Thus, discrepancies in search rates can arise even when law enforcers do not consider a person's race as relevant information in deciding whether to stop him.

A second, more nuanced, point made in the literature is that even if enforcers take race into account when making search decisions, this behavior does not necessarily imply racial prejudice, as the phrase is defined in prior work. Explaining this point further requires making a distinction between taste-based discrimination, which according to prior work constitutes racial prejudice, and statistical discrimination, which does not. Taste-based discrimination refers to instances where the enforcer has a preference towards mistreating, or at the very least showing less concern for, members of a particular race. Thus, taste-based discrimination, sometimes also called *racism*, occurs when enforcers attach different costs (or benefits) to searching members of different races. On the other hand, statistical discrimination occurs when law enforcers use a person's race as an input in evaluating the *likelihood* with which they may be engaging in criminal activity. It is important to note that an enforcer engaging in the latter type of discrimination is viewed as engaging in, loosely speaking, an innocent act, since his act is not motivated by a preference towards mistreating members of a certain race or some preference based racial bias. Moreover, statistical discrimination can help an enforcer maximize his rate of successful searches, because it increases the accuracy of the enforcer's predictions. Motivated by this distinction, a large body of literature focuses on empirically distinguishing

[1] *See, e.g.,* Starr (2015 p. 4) describing the differences in policing interactions across races.

statistical discrimination from taste-based discrimination which turns out to be a challenging task.[2]

An important question which is not raised frequently in the literature, however, is whether there is a good reason to rank one type of discrimination as better or worse than the other. Labeling one type of discrimination as racism and the other as statistical leaves the impression that the former must be worse than the latter. If the ultimate objective were to pass judgement on enforcers (who have racist preferences), as opposed to mitigating the social harms that may arise from discrimination, this type of ranking could perhaps be justified. There may also be other, moral, justifications for this type of normative ranking between statistical and taste-based discrimination, and a similar ranking can perhaps emerge if one ranks the two types of discrimination based on how often they lead to wrongful convictions.[3] The ranking may surprisingly be reversed, however, if one focuses on the impact of the two types of discrimination on crime. The implicit assumption that many scholars make is that statistical discrimination is also justified on grounds of enhancing deterrence, because it allows non-racist enforcers to maximize success rates. This deduction contains an error, because deterrence is (typically) not maximized by enforcement choices that maximize success rates.

This point, that maximizing the success rate does not necessarily maximize deterrence, has been crystallized in a line of research that investigates the optimal standard of proof and related issues (Schrag and Scotchmer (1994), Farmer and Terrell (2001), Demougin and Fluet (2006), Kaplow (2011a) and (2011b), and Lando and Mungan (2018)). An important take-away from this literature is that deterrence is a function of the gap between the expected net-benefits from committing crime versus refraining from crime (as in Png (1986), Schrag and Scotchmer (1994) and Polinsky and Shavell (2007)). This gap relates to the ex-ante incentives faced by an individual, which is shaped by the standard he expects enforcers to use in deciding whether to search him, and is otherwise unrelated to the criminal tendencies of the (racial) group to which he belongs. The 'success rate' frequently analyzed in the discrimination literature, however, is very sensitive to the crime rates among races, since it is essentially the posterior probability with which any given stopee is guilty. This probability naturally depends on the proportion of individuals who commit crimes, and approaches one [zero] as the crime rate approaches one [zero] within a given race. Thus, standards that maximize success rates depend on the criminal tendencies of the race a person belongs to, whereas the crime minimizing standard is independent of the same. Thus, eliminating discrimination can lead to the removal of a factor that ought to be irrelevant in determining the standards used in enforcement.

One may question how it is possible, then, for racial discrimination to have

[2] The roots of this literature can be traced back at least to Becker (1957), and this line of work continues to grow. Some of the most prominent and modern work is listed in Persico (2009), Antonovics and Knight (2009), and Anwar and Fang (2006).

[3] This type of analysis could be conducted in a framework where error costs (separate from deterrence and enforcement costs) are explicitly included in the social welfare function, see, e.g. Lando (2009), Demougin and Fluet (2005), and Chu et al. (2000).

ambiguous effects on crime rates in prior research (see, e.g. Persico (2002) and Bjerk (2007)). The most important reason for this ambiguity is that, absent discrimination, it is not clear whether enforcers would choose to use crime minimizing standards, because their incentives may not be aligned with crime minimization. When enforcers' incentives can be altered through rewards for accurate searches and liabilities for inaccurate searches, this ambiguity vanishes in many circumstances, because the misalignment in enforcers' incentives can be removed when they are unable to (or do not) engage in statistical discrimination. The implication is basically that when enforcer liabilities/rewards can be chosen appropriately, discrimination of all kinds, statistical as well as taste-based, result in an increase in crime.

These observations suggest that there is no obvious reason to think that statistical discrimination is less detrimental to the criminal justice system than taste-based discrimination. In fact, the analysis reveals that reductions in deterrence due to statistical discrimination are harder to eliminate than similar reductions caused by taste-based discrimination. This is because the internal costs/benefits that enforcers may incur due to their racist preferences can be fully diluted, in the limit, by using very large liabilities/rewards. However, the same method cannot be used to combat statistical discrimination, because these occur due to differences in the probabilities with which enforcers expect to get a 'hit' by stopping people of different races, as opposed to having different costs associated with stopping individuals of different races.

In short, this article illustrates that statistical discrimination can be as detrimental to deterrence as taste-based discrimination. This observation creates doubt about the supposition held by many that statistical discrimination is less bad than taste-based discrimination. However, it is quite important to note the exact scope of this observation and the conditions under which it is derived. Perhaps most importantly, like many prior articles, the observations made here relate *exclusively to deterrence*. They do not focus or rely on any other welfare or ethical implications of discriminatory practices, including deontological implications[4] and public outrage associated with racial discrimination; social costs associated with wrongful punishments[5]; changes in preventive benefits[6] and/or enforcement costs[7] due to discriminatory practices; and the impact of anti-discrimination policies on enforcer's well-being. Even this short list of potential considerations reveals how a welfare analysis will be sensitive to the specific

[4] *See, e.g.,* Lippert-Rasmussen (2006) discussing whether racial profiling is justifiable from a utilitarian as well as a deontological perspective.

[5] See note 3, above.

[6] Although harm prevention is an extremely important function that the criminal justice system seeks to further, it has not received nearly as much attention as deterrence in the theoretical economics of law enforcement literature. Friehe and Tabbach (2013) and Mungan (2018) are recent articles that provide a framework to incorporate the preventive benefits of law enforcement.

[7] It is also worth noting that the impact of discrimination on administrative costs are sensitive to which costs ought to be included in the analysis and to what extent. It is unclear, for instance, what fraction of the enforcer's utility from wrongful and correct stops ought to be included in the social welfare function when he is racist. It is similarly unclear what proportion of the disutility imposed upon a stoppee is transferable.

components included in the welfare function. Thus, the objective of this article is not to suggest that discriminatory practices cannot enhance some specifically defined notion of welfare, but to point out that there is no simple justification for such practices that are *based on the maximization of deterrence*.

A second important aspect of the analysis is that it derives the observation that discriminatory practices are detrimental to deterrence (in sections 3 and 4) under two race-neutrality assumptions that are also imposed in prior literature[8]: members of all races (i) face the *same cost ratio* associated with being investigated by enforcers when they are guilty versus innocent, and (ii) generate signals informing enforcers of the criminality of their behavior through processes that possess similar likelihood ratios. However, further analysis of the model (presented in section 5) reveals that discrimination continues to be detrimental to deterrence even when one relaxes the first assumption to allow the race being discriminated against to face greater relative costs associated with being wrongfully stopped than the race which is favored through discriminatory practices. This condition appears intuitive, especially if one expects the race that is discriminated against at the enforcement stage to be discriminated against in subsequent stages of the criminal justice system as well. These race-neutrality assumptions are explained in further detail after the model is introduced. However, it is worth highlighting here that within the confines of the model presented, it appears that one can support a claim that discriminatory practices can enhance deterrence only if one believes that the suspicion generation processes for different races greatly differ from each other in terms of their associated likelihood ratios. Otherwise, if one wishes to suggest that statistical discrimination may lead to benefits in the criminal justice system, they must be in the form of something other than enhancing deterrence, e.g. preventive benefits and/or administrative cost savings. Absent this type of justification, ceasing to distinguish between taste-based and statistical discrimination, and focusing on racial discrimination in law enforcement more generally, may facilitate the identification of problematic and deterrence-reducing enforcement practices. Moreover, simply communicating the negative impacts of statistical discrimination to the public as well as law enforcers may lead them to engage in such practices more carefully and selectively.

In the next section, I provide a brief literature review, which highlights the relationship between the findings of the instant article and prior scholarship on racial profiling. Subsequently, in section 3, I describe a simple model which is used to explain the interactions between law enforcers and potential criminals. Section 4 introduces enforcer liability/reward regimes, characterizes the optimal liability/reward regime, and explains how racial discrimination increases crime. Section 5 discusses some assumptions that are used in the analysis, and the effects of relaxing these assumptions. Section 6 contains concluding remarks, and an appendix in the end contains proofs of propositions.

[8] *See, e.g.,* Bjerk (2007). Schrag and Scotchmer (1994) invoke a similar assumption in a different context described in detail in section 2.

2 Literature Review

Becker (1957), Phelps (1972) and Arrow (1973) are pioneering theoretical works on discrimination. However, these articles, and a sizeable literature that builds on these articles (e.g. Coate and Loury (1993)), focus on the labor market. Nevertheless, this literature introduces concepts often used in the literature on police discrimination, including taste-based versus statistical discrimination.

A series of later articles, including Knowles et al. (2001), Persico (2002), and Hernandez-Murillo and Knowles (2004), incorporate statistical discrimination in a model where enforcers try to maximize the success rate among individuals they investigate. In these articles, police officers choose what proportion of each racial group to investigate. In equilibrium, the crime rate among investigated individuals in each racial group has to be equal. Otherwise, officers could increase the over-all success rate by increasing the rate at which they investigate individuals from the group for which the crime rate is higher. Thus, if the success rate is not equal across races, enforcers must be maximizing something other than success rates, which could include a 'weighted success rate' across racial groups, where weights are determined by officers' racial prejudices.

Based on this reasoning a very large body of literature has utilized success rates based empirical tests to identify taste-based discrimination. Moreover, a series of later articles, e.g. Persico and Todd (2005), (2006), and (2008), develop this literature further by explaining how the theoretical dynamics presented in Knowles et al. (2001) extend to more sophisticated frameworks. It is worth briefly noting that statistical discrimination is built into Knowles et al. (2001): Enforcers seek to maximize success rates, and, thus, officers' best responses to non-equilibrium strategy profiles where the crime rate differs across racial groups is to use different search rates across different races. In fact, the possibility of this type of statistical discrimination is what causes the equalization of crime rates and success rates in equilibrium.

Anwar and Fang (2006) and Bjerk (2007), in settings very similar to Coate and Loury (1993), have pointed out that enforcers can often base their search decisions on suspects' behavior which are affected by whether or not they are involved in criminal activity.[9] These behavioral differences can be modeled by assuming that criminals and non-criminals emit noisy signals regarding their involvement in criminal activity, and this allows law enforcers to update their beliefs regarding the likelihood with which suspects' are criminals based on these signals. In this framework, non-racist enforcers search a suspect whenever his posterior probability of being a criminal (i.e. the probability that the person is a criminal given his race and the signal he emits) is above a certain threshold, and this threshold can be race-dependent. Therefore, enforcers equate the success rate across the *marginal* offenders of different races. In other words, if it were possible to compare the success rates for stopees who have emitted the threshold signal applicable for their races, these success rates would have to be equal

[9] Dharmapala and Ross (2004) and Antonovics and Knight (2009) discuss other problems associated with the previously proposed game-theoretical framework by Knowles et al. (2001). These issues are briefly summarized in Anwar and Fang (2006 p. 132).

whenever enforcers are not engaging in taste-based discrimination.

Of course, empirically obtaining this type of information is difficult, if even possible. The type of information that can more easily be compared is the average success rates across races. Unfortunately, average success rates need not be equal across races, even when enforcers are not racially prejudiced, because different races may have different distributions of infra-marginal offenders (i.e. people who have been searched, because they have emitted signals that are above the threshold signal applicable for their race). This is a species of what is called the infra-marginality problem in the literature, which simply does not arise in earlier game theoretical models, because these models do not consider the possibility of enforcers basing their decisions on noisy signals emitted by suspects. Based on this observation, Anwar and Fang (2006) challenge the validity of using simple success rates tests to identify taste-based discrimination, and offer alternative tests.[10]

The instant article, like Persico (2002) and Bjerk (2007), assesses the likely effects of racial discrimination on criminal activity. However, unlike Persico (2002) this article considers a model where enforcers observe a noisy signal emitted by subjects which they can use to form suspicions, just as in Anwar and Fang (2006) and Bjerk (2007).[11] Moreover, unlike Persico (2002) and Bjerk (2007), the article explicitly incorporates taste-based discrimination as well as statistical discrimination, considers endogenously determined enforcer liabilities and benefits,[12] and incorporates the possibility of erroneous investigations causing investigatees harms. These additional features of the model are used to illustrate that discrimination, regardless of its type, increases crime. These results differ from the less determinate results obtained in Persico (2002) and Bjerk (2007), which both contain conditions relating to the distribution of benefits from (or opportunity costs to) committing crime among the two populations.

An additional, but perhaps less important feature of the model that distin-

[10] Although these tests are not central to the analysis proposed in the instant article, they can briefly be characterized as exploiting the presence of racial heterogeneity among enforcers, and the ranking of success rates across races based on officers' races. As Anwar and Fang (2006 p. 131) point out, their test "can detect only what we term to be *relative* racial prejudice, and not *absolute* racial prejudice. This is because when the ranking of search rates and search success rates over officer races depends on the race of the motorists, we know that at least one of the racial groups of officers is using racial prejudice, but we cannot identify which group it is. Thus, all we can conclude is that one group of troopers is more racially prejudiced relative to another group of troopers, instead of an absolute conclusion which would identify which groups of troopers were racially prejudiced."

[11] Both the instant model, as well as Anwar and Fang (2006) and Bjerk (2007), follow a modeling approach very similar to that in Coate and Loury (1993), which considers a setting where employers choose whether to make job-related investments and employers decide whether to assign employers to 'good' or 'bad' tasks based on a noisy signal emitted by the employee regarding his qualifications. Here, the decision to commit crime is analogous to the employee's investment decision, and the enforcer's searching decision is analogous to the employer's assignment decision.

[12] In section VIII.B. Persico (2002) considers the possibility of there being greater rewards to enforcers from conducting successful searches of members of one of the races than another. However, this analysis does not consider the possibility of using rewards and liabilities as policy tools.

guishes it from prior models is that it is capable of producing multiple equilibria (like in Coate and Loury (1993)). Thus, the model can be used for describing all three primary sources of discrimination considered in the literature (and combinations thereof): taste-based discrimination, statistical discrimination due to the presence of multiple equilibria where members of different races behave differently despite having identical characteristics, and statistical discrimination due to races having different characteristics.

Finally, there is a strand of literature that questions how various rules of evidence can be used to reduce decision makers' bias in deciding cases (e.g. Daughety and Reinganum (2000) and Schrag and Scothmer (1994)). The effect of rules considered in this literature is akin to reducing discrimination, since they result in treating like cases (and defendants) alike. Schrag and Scothmer (1994) is most illustrative of the similarities between these two strands of the literature. Schrag and Scothmer focus on the admissibility of character evidence and how it relates to the maximization of deterrence. As in the current framework, the authors note that deterrence is maximized when the gap between the returns from committing crime and remaining innocent is maximized. In the most relevant setting considered by Schrag and Scotchmer, juries are prejudiced against habitual offenders. Thus, prohibiting character evidence forces the jury to use a single standard in judging habitual and non-habitual criminals alike. This makes the jury less lenient towards non-habitual offenders, and more lenient towards habitual offenders, relative to the case where character evidence is allowed. This enhances deterrence when juries are assumed to have preferences over the wrongful convictions and acquittals of habitual and non-habitual offenders, respectively, which cause them to be too harsh, and too lenient, in judging these two respective groups.[13] In the instant framework, the police engage in racial discrimination in a similar manner. However, since it is impossible to prohibit officers from perceiving suspects' races, the policy tools considered here are liability and reward regimes instead of the complete elimination of discrimination. Perhaps counter-intuitively, these tools are more effective than those that categorically make certain types of evidence inadmissible, because they can be used even absent discrimination to align the decision maker's incentives with the objective of crime minimization. This is why the elimination of discrimination has ambiguous impacts on deterrence in the prior literature focusing on the hypothetical elimination of discrimination (e.g. Persico (2002) and Bjerk (2007)), but more clear implications in the current setting where rewards and penalties are available.

3 Model

In the interactions described below, each individual chooses whether or not to initiate a criminal act. Subsequently, each individual emits a noisy signal regarding his guilt, which is observed by a law enforcer. The enforcer chooses

[13] When there is no prejudice in the judicial system, Schrag and Scotchmer (1994) find ambiguous results, similar to those of Persico (2002) and Bjerk (2007) discussed, above.

the threshold suspicion that he uses to stop and search individuals. Searches can lead to punishment or other inconveniences for the stopees, and these generate costs which are, in expectation, smaller for innocent individuals than for guilty individuals.[14] Thus, as in Png (1986) and the subsequent large body of literature incorporating enforcement errors, the gap between the expected returns from committing crime and remaining innocent affects the incentives of potential offenders.[15] Therefore, the threshold suspicion used by law enforcers affects deterrence, and, deterrence, in turn affects enforcers' interpretation of signals.

To formalize discrimination, the model focuses on a population which consists of black (B) and white (W) individuals. Throughout the analysis, I use the letter R to denote race, and unless indicated otherwise it should be assumed that expressions which refer to R are valid for all $R \in \{B, W\}$. Without loss of generality, I assume that B is the race that is subject to either statistical, taste-based, or both types of discrimination.

The interactions between enforcers and individuals can be conceived of as a two period game. In the first period, individuals of both races simultaneously decide whether or not to engage in criminal activity. In the second period, enforcers choose -potentially different- threshold suspicions used to stop black and white individuals. Depending on whether or not enforcers observe the crime rate among the two races prior to making their decisions, the game can be analyzed as a simultaneous or sequential move game. However, as will become clear from the proceeding analysis, the two versions of the game produce the same subgame perfect equilibria. Therefore, I present the interactions between enforcers and the population as a sequential move game to ease the description of the model, and I start by deriving the best responses of law enforcers to crime rates, and subsequently characterize equilibria.

3.1 Law Enforcers' Decision Making Process

When a law enforcer encounters a suspect, he must choose whether or not to stop him. He faces costs and benefits from conducting wrongful and correct stops, respectively. In particular ψ_R denotes the cost of incorrectly stopping an innocent individual[16] of race R, and π_R denotes the benefit from correctly stopping a guilty individual of race R (both, relative to not stopping him). These values include the effort that goes into stopping an individual, internal psychic costs and benefits, and the impact stops may have on an enforcer's career development. However, enforcer liabilities and rewards that may be used to alter these values are ignored until section 4. Thus, if q is the probability with which a given suspect is guilty, the enforcer's expected net benefit from

[14] Here, and in the remaining parts of the article, individuals who engage in criminal acts are called guilty, and those who refrain from such acts are called innocent.

[15] This literature includes Garoupa and Rizzolli (2012), Nicita and Rizzolli (2014), and Polinsky and Shavell (2007), and often credits Png (1986) for having first emphasized the point that wrongful punishments reduce deterrence (see, e.g., Polinsky and Shavell (2007) n. 43 and accompanying text).

[16] One can, instead, use the cost of conducting *any* investigations, rather than the cost of conducting inaccurate stops. This has no effect on the results presented.

stopping the person is $q\pi_R - (1-q)\psi_R$, which implies that the enforcer conducts a stop if

$$q_R \equiv \frac{\psi_R}{\pi_R + \psi_R} \leq q \tag{1}$$

Here, q_R can be thought of as the threshold suspicion that triggers a stop. The subscript R reflects the fact that this threshold can be race dependent, which would cause taste-based discrimination. To incorporate this possibility, it is assumed that $\psi_B \leq \psi_W$ and $\pi_B \geq \pi_W$, since then $q_B \leq q_W$.

In estimating a suspect's probability of guilt, q, enforcers rely on a noisy signal, $x \in [\underline{x}, \overline{x}]$ emitted by the individual, as an input. Functions f_0 and f_1, respectively, denote the continuous densities of the signal when the person is guilty versus when he is innocent with the corresponding CDFs denoted F_0 and F_1.[17] Thus, the probability with which a suspect of race R who emits signal x is guilty is given by:

$$q(\theta_R, x) = \frac{\theta_R f_0(x)}{\theta_R f_0(x) + (1-\theta_R) f_1(x)} \tag{2}$$

where θ_R is the crime rate determined by the joint choices of all individuals of race R in period 1. This expression highlights that enforcers are assumed to engage in statistical discrimination, whenever possible, since they take race as an input in determining q. The contrary assumption is considered only when the effects of statistical discrimination are analyzed in sections 4 and 5.

When the signal generating process, summarized by f_0 and f_1, satisfies the Monotone Likelihood Ratio Property (MLRP) the threshold probability described in (1) corresponds to a threshold signal emitted by suspects. In particular, when $\frac{d(\frac{f_0}{f_1})}{dx} > 0$, it follows that large x are more consistent with guilt. If in addition, there exist conclusive signals of guilt, i.e. $f_0(\overline{x}) > f_1(\overline{x}) = 0$ and $f_1(\underline{x}) > f_0(\underline{x}) = 0$, then for extreme signals enforcers can be almost certain of suspects' innocence or guilt.[18]

Given these assumptions, for all crime rates θ_R and each race, there exists a unique threshold signal

$$\widehat{x}_R \in (0,1) \text{ such that } q(\theta_R, \widehat{x}_R) = q_R \tag{3}$$

because

$$q(\theta_R, 0) = 0,\ q(\theta_R, 1) = 1, \text{ and } \frac{\partial q(\theta_R, x)}{\partial x} > 0 \text{ for all } \theta_R, x \in (0,1) \tag{4}$$

where the last property follows from MLRP. Therefore, an enforcer's best response, previously expressed in (1), is to stop an individual of race R whenever he emits a signal $x \geq \widehat{x}_R(\theta_R, q_R)$. To make easy references to stopping rules that

[17] Following Coate and Loury (1993), these distributions are assumed to be race-independent. The implications of relaxing this assumption are discussed in section 5.

[18] A similar assumption is invoked by Anwar and Fang (2006).

enforcers may employ in equilibrium, I refer to such stopping rules as *threshold rules*.

It easily follows that enforcers' best responses satisfy the following properties:

$$\frac{\partial \widehat{x}_R(\theta_R, q_R)}{\partial \theta_R} < 0; \lim_{\theta_R \to 0} \widehat{x}_R(\theta_R, q_R) = 1; \lim_{\theta_R \to 1} \widehat{x}_R(\theta_R, q_R) = 0; \text{ and } \frac{\partial \widehat{x}_R(\theta_R, q_R)}{\partial q_R} > 0 \tag{5}$$

These properties imply that when $q_W > q_B$, in equilibrium, enforcers will employ a lower standard while stopping black individuals, if the two races have the same crime rate. Similarly, if, in equilibrium, the crime rate is higher among black individuals, enforcers will employ a lower standard in stopping them, even if they do not engage in taste-based discrimination.

3.2 Individuals' Decision Making Process

All individuals have the option of committing crime and receiving criminal gains of γ.[19] These gains vary across individuals and the cumulative distribution function (CDF) that describes the gains within the black and white population are denoted as G_B and G_W, respectively, with $g_R \equiv \frac{dG_R}{d\gamma} > 0$ with support $[0, \infty)$. To incorporate the possibility of statistical discrimination -in the absence of multiple equilibria- I assume that G_B may first order stochastically dominate (FOSD) G_W, i.e. $G_B(\gamma) < G_W(\gamma)$ for all $\gamma \geq 0$. But, to allow for discrimination that is either exclusively taste-based, or is a result of the presence of multiple equilibria, I also consider cases where G_B and G_W are identical.

Given any threshold rule x_R, it follows that the probability with which guilty and innocent individuals are stopped are given by

$$\beta_R \equiv \beta(x_R) \equiv 1 - F_0(x_R); \text{ and} \tag{6}$$

$$\alpha_R \equiv \alpha(x_R) \equiv 1 - F_1(x_R), \tag{7}$$

respectively. As the analysis in the previous sub-section demonstrates, threshold rules are the only types of strategies that can be employed by enforcers in equilibrium, thus, for purposes of identifying equilibria, these are the only relevant stopping rules.

Guilty and innocent people who are stopped face expected costs of c_0 and c_1, respectively.[20] Thus, for a person of race R, the expected pay-offs from

[19] It is worth noting that this analysis relies on Becker's (1968) standard and purely simplifying assumption that the criminal gains, γ, are obtained as soon as one commits the crime. This means that the criminal obtains these benefits, even if he is searched, i.e. criminal benefits are not forefeited upon a search. (See, e.g., Shavell (1990), for a very similar simplifying assumption where individuals receive criminal benefits even if their attempts are unsuccesful) Results are preserved under a broad set of more realistic circumstances where (i) only a proportion of criminal benefits are obtained prior to the (potential) search *or* (ii) searches lead to the forefeiture of illegal gains with a probability less than one. These assumptions can be incorporated by using a modified version of the Beckerian model used in Mungan and Klick (2014) which requires the introduction of additional notation.

[20] Note that these costs can be race-dependent. This possibility is considered in section 5.

committing and refraining from committing crime are $\gamma - \beta_R c_0$ and $-\alpha_R c_1$, respectively. Therefore, a person commits crime if:

$$\widehat{\gamma}(x_R) \equiv \beta_R c_0 - \alpha_R c_1 < \gamma \tag{8}$$

Here $\widehat{\gamma}(x_R)$ summarizes a critical value that an individuals' criminal benefit must exceed to make committing crime the more profitable option. This critical value is a function of the threshold rule x_R, which affects the gap between the expected costs associated with committing crime and remaining innocent through its impact on the stop probabilities β_R and α_R defined in (6) and (7). Thus, the level of deterrence generated by individuals' best responses is given by $G_R(\widehat{\gamma}(x_R))$, and its properties are summarized by the following lemma.

Lemma 1 *(i) The level of deterrence generated by the best responses of individuals of race $R \in \{B, W\}$ is single peaked in the threshold rule x_R, and (ii) there exists a threshold rule $x^m \in (0,1)$ such that* $\arg\max_{x_B} G_B(\widehat{\gamma}(x_B)) = \arg\max_{x_W} G_W(\widehat{\gamma}(x_W)) = x^m$.

Proof. Differentiating $\widehat{\gamma}$ with respect to x_R reveals that:

$$\frac{d\widehat{\gamma}}{dx_R} = f_1(x_R)c_1 - f_0(x_R)c_0 \geq 0 \text{ iff}$$

$$L(x_R) \equiv \frac{f_0(x_R)}{f_1(x_R)} \leq \frac{c_1}{c_0} \tag{9}$$

L, which denotes the likelihood ratio, is increasing in x_R due to MLRP, with $L(\underline{x}) = 0$, and $\lim_{x_R \to \overline{x}} L(x_R) = \infty$. Thus, there exists x^m such that

$$L(x^m) = \frac{c_1}{c_0} \tag{10}$$

Therefore, $\frac{d\widehat{\gamma}}{dx_R} \geq 0$ iff $x_R \leq x^m$. ■

Part (i) of Lemma 1 reveals that deterrence is increasing in the threshold rule only up to a point (i.e. x^m). This non-monotonicity result is analogous to those that emerge in the prior literature (e.g. Demougin and Fluet (2006) and Schrag and Scotchmer (1994)). The result is a direct consequence of the fact that using weaker standards for conducting stops increases the probability of wrongful as well as correct stops. Moreover, as the standard is made weaker, its impact on the probability of wrongful stops relative to its impact on the probability of correct stops (i.e. f_1/f_0) becomes larger, since weak signals are more often produced by innocent individuals than guilty individuals (which is reflected by the assumption of MLRP). Thus, when the threshold rule is sufficiently weak (i.e. weaker than x^m) making it even weaker causes the gap between the expected costs associated with committing crime and remaining innocent to shrink, and this reduces deterrence.

Moreover, Lemma 1 implies that among all threshold rules, x^m maximizes the deterrence of black as well as white individuals. The reason for this is

that deterrence is positively related to $\widehat{\gamma}$, which is the minimum criminal gain a person must have to commit crime. This critical benefit is race-independent, because its determinants (namely the likelihood ratio, and the cost ratio ($\frac{c_1}{c_0}$)) are assumed to be race-independent (the implications of relaxing this assumption are considered in section 5). It is worth noting that, despite having identical maximizers, the *level of deterrence* can be different across the two races. This is because the level of deterrence is measured by the proportion of individuals who choose not to commit crime (i.e. $G_R(\hat{\gamma}(x))$, and the two races may have different proportions of individuals who have criminal benefits above a given criminal benefit threshold. This can be noted, for instance, by observing that when G_B FOSD G_W it follows that $G_B(\hat{\gamma}(x^m)) < G_W(\hat{\gamma}(x^m))$.

3.3 Equilibrium Characterization

An equilibrium is obtained when no party can profitably deviate from his/her strategy given all other parties' strategies. To characterize equilibria, it is useful to define the following probability of guilt:

$$Q_R(x_R) \equiv q(\widehat{\theta}_R(x_R), x_R) \text{ for } R \in \{B, W\} \text{ where} \tag{11}$$

$$\widehat{\theta}_R \equiv 1 - G_R(\widehat{\gamma}(x_R)) \tag{12}$$

Here Q corresponds to the likelihood with which a person who emits signal x_R is guilty, when people of race R are playing their best responses to the threshold rule x_R, and, hence, produce a crime rate of $\widehat{\theta}_R$. Thus, the enforcer's strategy, summarized by the threshold rule x_R^* such that

$$Q_R(x_R^*) = q_R \tag{13}$$

along with individuals' strategies of committing crime only if $\gamma > \widehat{\gamma}(x_R^*)$ constitute an equilibrium.

That an equilibrium exists follows from the simple observation that $Q_R(\underline{x}) = 0 < q_R < Q_R(\overline{x}) = 1$ for all $q_R \in (0, 1)$. Moreover, multiple equilibria can exist, since

$$Q_R' = \frac{\partial q}{\partial \theta_R}\frac{d\widehat{\theta}_R}{dx_R} + \frac{\partial q}{\partial x_R} = -g_R(\widehat{\gamma}(x_R))\frac{\partial q}{\partial \theta_R}\frac{d\widehat{\gamma}}{dx_R} + \frac{\partial q}{\partial x_R} \tag{14}$$

and, as noted in lemma 1, $\frac{d\widehat{\gamma}}{dx_R} > 0$ for all $x < x^m$. Thus, it is possible for Q_R to be decreasing in some intervals within $(0, x^m)$, because $g(\widehat{\gamma})$ can be large enough to cause this result.[21] This possibility is depicted in figure 2, below, which illustrates how multiple equilibria can emerge. To simplify exposition, the figure focuses on potential multiple equilibria when blacks and whites have identical criminal benefit distributions, i.e. $G_B \equiv G_W$, and when there is no taste-based discrimination, i.e. $q_B = q_W$.

It is also worth highlighting that $Q_B(x) > Q_W(x)$ for all $x \in (0, 1)$ whenever G_B FOSD G_W, because then $\widehat{\theta}_B(x) > \widehat{\theta}_W(x)$. Figure 3, below, makes use of this observation.

[21] A complete proof of this statement is omitted, because similar results have been presented in the literature, see e.g. Rasmusen (1996), Coate and Loury (1993), or Mungan (2016).

3.4 Emergence of Discrimination due to Different Reasons

The equilibrium characterization in (13) is useful for discussing how discrimination may emerge due to three different reasons. Among the three sources of discrimination, the presence of racist preferences is the simplest to explain: if enforcers perceive different costs to incorrectly stopping members of different races (or alternatively benefits to correctly stopping members of different races), then they will use different suspicion thresholds (i.e. q_R) in stopping people. This, in turn, can cause them use different threshold rules in stopping members of different races. This possibility is depicted in figure 1, below, which considers a case where G_B and G_W are identical to isolate the effect of taste-based discrimination alone, and where Q_R is upward sloping throughout, to eliminate the possibility of multiple equilibria.

[Insert Figure 1, here]

As noted via (14) multiple equilibria can exist in cases where changes in the stopping standard cause large impacts on the crime rate. In these cases, even when the two races have identical criminal propensities, one can be subjected to a higher standard than the other. This possibility is depicted in figure 2, below, which focuses on equilibria represented by the points E_1, E_2 and E_3, where only E_1 and E_3 are stable. These two equilibria are associated with threshold rules $x_1 < x_3$, where x_3 represents a stronger standard, since enforcers stop individuals only upon the receipt of stronger signals. In this example, black individuals will be subject to a weaker standard, if, for instance, they adopt a strategy of committing crime when they have criminal benefits that exceed $\widehat{\gamma}(x_1)$, while white individuals commit crime only if their benefits exceed $\widehat{\gamma}(x_3) > \widehat{\gamma}(x_1)$. This corresponds to a case where the discrimination is caused by the two groups' different responses to identical circumstances, and enforcers' response of using this information (which is conveyed to them through suspects' races) to statistically discriminate when choosing threshold rules. The interpretation of this result changes to some extent when enforcers are assumed to act without knowledge of groups' crime rates and only form beliefs about these crime rates. In such circumstances, the existence of multiple equilibria can be interpreted as being due to *self-fulfilling expectations*: enforcers' beliefs influence the equilibrium standards and crime rates.[22] Regardless of the interpretation, discrimination that emerges exclusively due to the presence of multiple-equilibria are caused by enforcers using race as an input in assessing suspects' likelihood of guilt.

[Insert Figure 2, here]

The last source of discrimination considered is differences across the characteristics of the two races, which are captured by differences in G_B and G_W. In these cases enforcers associate different probabilities of guilt with individuals of different races who produce the same signal, due to the differences in the crime rates across the two races. Thus, differences in the equilibrium threshold

[22] *See, e.g.,* Rasmusen (1996) for a similar interpretation.

standard are caused due to differences in Q_B and Q_W, as opposed to q_B and q_W (which is the cause of taste-based discrimination). This possibility is depicted in figure 3, below.

[Insert Figure 3, here]

These three possibilities clarify the terms used to describe various sources of discrimination. The next proposition formalizes the idea that the presence of discrimination can be caused by any of these three sources, and also specifies the relationship between the equilibrium standards.

Proposition 1 *(i) When there is a single equilibrium, enforcers use lower standards in stopping black suspects (i.e. $x_B^* < x_W^*$) if $q_B < q_W$; G_B FOSD G_W; or both. (ii) Moreover, law enforcers may use lower [higher] standards in stopping black suspects (i.e. $x_B^* < x_W^*$ [$x_B^* > x_W^*$]), even when G_B and G_W are identical and $q_B = q_W$, if there exist multiple equilibria.*

Proof. See Appendix. ■

As the proposition demonstrates, discrimination can be caused by the presence of multiple equilibria, or a combination of the two remaining sources of discrimination. In the remaining parts of the analysis, I focus on cases where there is a single equilibrium. In addition to simplifying the analysis, this approach can be justified by noting that eliminating discrimination caused by multiple equilibria requires policies that are different, in nature, than those required to eliminate discrimination caused by the other two sources. In particular, multiple equilibria can potentially be eliminated by solving the coordination problem among individuals of the race who choose strategies leading to the higher crime rate. Similarly, if the existence of multiple equilibria is caused by enforcers having different beliefs regarding the two races, providing small compensations to police departments that are inversely related to the crime rate in their districts may cause them to commit to the stopping standard that leads to the highest level of deterrence. However, problems caused by taste-based discrimination and the presence of different racial characteristics cannot be solved by a tilting of the equilibrium towards another one that leads to less crime.

Thus, in the remainder of the analysis, I focus on cases where these is a single equilibrium, which also allows references to the two remaining sources of discrimination as taste-based and statistical, without causing any ambiguity.

4 The Impact of Discrimination on Crime with Enforcer Liability

The preceding sections took the cost and benefits to law enforcers from conducting stops as given, and focused on making observations regarding crime rates and standards employed. This section first introduces a liability/reward regime for law enforcers, which, in effect, endogenizes q_B and q_W. Then, it demonstrates that both statistical and taste-based discrimination increase crime. Subsequently, it derives crime minimizing liability/reward regimes.

4.1 Enforcement Liability Regimes

To abbreviate descriptions, I will refer to all regimes as *liability regimes*, although I will consider both sticks and carrots as potential policy variables to influence enforcer behavior. In particular, I consider regimes that impose a liability of λ_R on law enforcers for conducting wrongful stops, but which provide them with rewards of μ_R for conducting accurate stops. These regulatory liabilities and rewards are in addition to enforcers' internal costs from wrongful stops (of ψ_R) and internal benefits from accurate stops (of π_R). Thus, the total cost of a wrongful stop is $\psi_R + \lambda_R$, and the total benefit from an accurate stop is $\pi_R + \mu_R$. The R sub-scripts reflect the fact that enforcers may continue to engage in taste-based discrimination, and, at least theoretically, liabilities and rewards can be based on the stopee's race. To highlight the latter point, one can distinguish between two types of liability regimes as follows. The introduction of λ_R and μ_R makes the notation slightly more cumbersome, but is necessary to distinguish between race-neutral and race-dependent liability regimes, which are concepts defined as follows.

Definition 1 *A liability regime is race-neutral if $\lambda_B = \lambda_W$ and $\mu_B = \mu_W$. Liability regimes that are not race-neutral are called race-dependent.*

4.2 Impact of Discrimination on Crime

In order to formalize the impact of discrimination on crime, it is useful to consider the hypothetical case where enforcers are color-blind, in the sense that they cannot (or do not) take a person's race into account when deciding whether or not to stop him. In this case, enforcers do not form separate expectations regarding the crime rate among the two races, but, must form expectations regarding the aggregate crime rate. To formalize this idea, let $G_N \equiv \phi G_B + (1-\phi)G_W$ denote the CDF describing the criminal gain distribution among the entire population, where ϕ is the proportion of blacks in the population. Moreover, let q_N and Q_N refer to the analogs of q_R and Q_R in the previous section. It immediately follows that by choosing a (race-neutral) liability λ and reward μ, such that $q_N(\lambda, \mu) = Q_N(x^m)$, one can minimize the crime rate among both races. In this case, the total crime rate, denoted Θ, is minimized at

$$\Theta^m \equiv \phi\theta_B^m + (1-\phi)\theta_W^m \tag{15}$$

where

$$\theta_R^m \equiv 1 - G_R(\widehat{\gamma}(x^m)) \tag{16}$$

is the lowest crime rate achievable for race R. An immediate corollary of this observation is that discrimination increases crime.

Proposition 2 *Discrimination leads to more crime than Θ^m.*

Proof. See Appendix. ■

It is worth clarifying that proposition 2 refers to discrimination generally, which, in turn could be taste-based discrimination, statistical discrimination, or a combination of the two types of discrimination. Thus, when liability regimes can be used to align non-discriminating enforcers' incentives with the social objective, it unambiguously follows that discrimination increases crime.

4.3 Elimination of Discrimination through Liability Regimes

A remaining question is whether, when enforcers are not color-blind, their discriminatory behavior can be neutralized through appropriate liability regimes. The next proposition answers this question.

Proposition 3 *Suppose that discrimination occurs in the absence of a liability regime, then:*

Discrimination can be eliminated (and a crime rate of Θ^m can be achieved) only through race-dependent liability regimes which impose greater liability on law enforcers for wrongfully stopping members of the race being discriminated against (i.e. $\lambda_B > \lambda_W$) and/or providing smaller rewards to accurately stopping members of the race being discriminated against (i.e. $\mu_B < \mu_W$). Thus, it is impossible to eliminate discrimination through race-neutral liability regimes.

Part (ii) of the proposition demonstrates that, when race-dependent liability regimes can be used, discrimination can be countered through liability regimes which impose greater liability for wrongful stops of black individuals compared to white individuals. These race-dependent policies are, themselves, racially discriminatory and may therefore be impermissible. However, the observation may be used for discussing the desirability of any *de facto* asymmetric treatment of police misconduct which victimizes black individuals versus white individuals. If, for instance, (social) media pays disproportionate attention to cases where the victim of police misconduct is black versus white, and this leads to unequal expected liabilities imposed on enforcers as a function of the victim's race, this *may* have a subtle and positive effect on deterrence. However, if these effects are small and race-dependent liability regimes are impermissible, it becomes impossible to eliminate discrimination as pointed out by part (i), and this is detrimental to deterrence, as pointed out by proposition 2. In such cases, one can investigate the properties of the race-neutral crime minimizing liability regime.

4.4 Crime Minimizing Race-Neutral Liability Regimes

When discrimination occurs in the absence of a liability regime, and liability regimes are constrained to be race-neutral, it is impossible to eliminate discrimination. Nevertheless, one can design liability regimes to minimize the impact of discrimination on the total crime rate. The next proposition describes the characteristics of liability regimes that achieve this goal.

Proposition 4 *(i) When enforcers engage in statistical, but not taste-based, discrimination, the race-neutral liability regime that minimizes total crime results in a standard for white [black] people that is higher [lower] than x^m. The*

resulting crime rates, denoted $\theta^P_{R\in\{B,W\}}$, are both greater than the minimum crime rates achievable among the two races in the absence of discrimination.

(ii) When enforcers engage in taste-based discrimination, there exists no finite liability/reward combination that minimizes the crime rate, because the crime rate can always be decreased further by appropriately and simultaneously increasing λ and μ. In particular, there exists a function $\widehat{\mu}(\lambda)$, such that $\lim_{\lambda\to\infty} \theta^*_{R\in\{B,W\}}(\lambda, \widehat{\mu}(\lambda)) = \theta^P_{R\in\{B,W\}}$. This means that the adverse effects on crime rates due to taste-based discrimination can be completely eliminated in the limit.

Proof. See Appendix ■

Two interesting implications of proposition 4 are worth highlighting. First, the increases in crime resulting from taste-based discrimination can be eliminated almost completely through liability regimes that impose large costs for wrongful stops along with large rewards for accurate stops. The same is not true for problems arising due to statistical discrimination. Thus, rather counter-intuitively, statistical discrimination poses a greater problem compared to taste-based discrimination in this respect. Second, if the crime minimizing liability regime is implemented, in equilibrium, the standard faced by black individuals is weaker than the crime minimizing standard. This means that increasing the rate of type-1 errors faced by these individuals (through a weakening of the standard x_B) reduces deterrence, and the opposite result holds for white individuals. This last observation relates to the literature on the effect of type-1 errors on deterrence,[23] and suggests that type-1 errors may have asymmetric effects on the deterrence of members of different races.

In interpreting part (ii) of proposition 4, it is important to note that, in reality, there will be an upper bound to λ beyond which enforcers cannot be made to incur the costs of liabilities due to problems of judgment-proofness. Thus, the adverse impacts of taste-based discrimination on deterrence cannot be completely eliminated. However, a simple corollary of the limit result reported is that when the maximum liability that can be imposed upon enforcers is sufficiently large[24], increases in crime caused by statistical discrimination are greater than the increases in crime caused by taste-based discrimination when the liability regime is chosen optimally. This result is in tension with the impression created in the literature that statistical discrimination is socially less detrimental than taste-based discrimination.

[23] *See, e.g.,* Png (1986), Lando (2006), Garoupa and Rizzolli (2012), and Lando and Mungan (2018).

[24] One method to increase the effectiveness of the liability regime would be to condition the liability or reward on the stopping rule of the enforcer, instead of the result of the stop, as has been assumed throughout. However, since a decision maker cannot directly observe the stopping rule used by the enforcer, this is not a feasible method to increase the maximum effective liability. Nevertheless, the maximum liability can be significantly increased through non-monetary sanctions, e.g. delayed or reduced prospects for promotion, or access to enforcers' pension funds.

5 On the Generality of the Result that Discrimination Increases Crime

The analysis in the previous section makes a number of natural assumptions to analyze the impact of discrimination on crime. In particular, the analysis assumes that the expected costs faced by guilty and innocent individuals (i.e. c_0 and c_1), as well as the signal generation processes (i.e. f_0 and f_1), are race independent. Thus, the condition that characterizes the crime minimizing standard for both races, previously defined as x^m is characterized by the following equality:

$$L(x^m) = \frac{f_0(x^m)}{f_1(x^m)} = \frac{c_1}{c_0} \tag{17}$$

where L refers to the likelihood ratio. When either of these assumptions is relaxed, the crime minimizing standard may differ across the two races. These possibilities can easily be studied by focusing on a modification of (17) to allow for the possibility of both the likelihood ratio being race dependent and what I call the *cost ratio*, which refers to $\frac{c_1}{c_0}$, to be race dependent. In this case, the potentially race-dependent crime minimizing standards are characterized by:

$$C_R \equiv \frac{c_1^R}{c_0^R} = L_R(x_R^m) \tag{18}$$

Here, both C_R and L_R, respectively, refer to the cost and likelihood ratios for race R, and x_R^m refers to the potentially race-dependent crime minimizing threshold rule. When the cost and likelihood ratios are race-dependent, one may wonder under what circumstances the result pertaining to the impact of discrimination on crime is preserved. The next lemma identifies simple sufficient conditions under which discrimination increases crime.

Lemma 2 *Whenever $x_B^m \geq x_W^m$ and enforcer liability is chosen to minimize crime, discrimination against blacks increases crime.*

Proof. See Appendix. ■

The observations made in lemma 2, above, do not relate to the primitives of the model, but, instead refer to the relationship between the crime minimizing threshold rules, which are determined by the primitives of the model. However, it provides a very useful starting point for identifying conditions under which discrimination continues to increase crime. Specifically, it notes that this result holds when the crime minimizing standard for black individuals is stronger than the crime minimizing standard for white individuals. This condition, in turn, holds when either $C_B \geq C_W$, or when the signals generated by black individuals are more informative in a particular sense described, below.

Considering first the case where the cost ratio is greater for black individuals, note that since $L'_R > 0$ it immediately follows that $x_B^m > x_W^m$ when the likelihood ratios are race-independent (i.e. $L_B \equiv L_W$), but $C_B \geq C_W$. This corresponds to a case where the expected punishment black people face upon

being wrongfully stopped, normalized by the expected punishment they face when they are stopped when guilty, is greater than the analogous ratio faced by white individuals. This is likely if discriminating enforcers tend to fabricate evidence more against people who belong to the race that they discriminate against. This possibility is depicted in figure 4, below.

[Insert Figure 4, here]

Next, consider the case where signals produced by the black population are, in a particular sense, more informative. Although more rigorous formalizations are possible, for purposes of this discussion, this point can be demonstrated by analyzing a situation where $L_B(\widehat{x}) = L_W(\widehat{x}) = 1$ for some $\widehat{x} \in (0, 1)$, and $L_B(x) \leq L_W(x)$ iff $x \leq \widehat{x}$ (see figure 5, below). In this case, it follows that the signals produced by the black population are more informative, because they allow for better discrimination between innocent and guilty individuals. In this case, even when $C_B = C_W$, the crime minimizing standard for the black population is higher. This is depicted in figure 5 below. This conclusion naturally also holds when the cost ratio for blacks is greater. This case is also depicted in figure 5 with gray lines and letters.

[Insert Figure 5, here]

Among these two potential sources of heterogeneity across races, the one that appears to be harder to interpret relates to variations in the signal generating processes. It is hard to see why the race that is being discriminated against may have a significantly more or less informative signal generating mechanism. Therefore, the interpretation of possibilities relating to different signal generating processes is left outside the scope of this article. What can be pointed out, however, is that as long as black individuals face greater relative costs associated with being wrongfully stopped than white individuals, the crime minimizing threshold rule is stronger for black individuals than white individuals, provided that the signal generating processes across the two races are not much different than each other (in either direction). Moreover, this conclusion continues to hold, even when the signal processes are significantly different than each other, so long as the signals emitted by black individuals are, loosely defined, more informative. In all of these circumstances, discrimination has a deterrence reducing effect.

Finally, it is worth pointing out that this sub-section has focused exclusively on sufficient conditions under which discrimination is likely to increase crime, and, thus, the set of circumstances under which this conclusion holds is naturally broader. For instance, as can be inferred from the logic behind the proof of lemma 2, even in cases where x_W^m is slightly larger than x_B^m, discrimination can cause an increase in crime, if it generates gaps between the threshold rules applicable to the two races that are sufficiently wider than the gap between x_W^m and x_B^m.

6 Conclusion

Scholars have made a considerable effort to distinguish between taste-based and statistical discrimination. Some of the discussions in this line of work leaves the impression that there is an understanding among many scholars that statistical discrimination is somehow less harmful than taste-based discrimination. This article challenges this understanding, and suggests that, *a priori*, there is no reason to think that statistical discrimination promotes the goals of the criminal justice system. In fact, it demonstrates that, when enforcer's incentives can be altered through liabilities and rewards, deterrence is unambiguously reduced as a result of discrimination. In concluding, it is worth pointing out that, although deterrence is very frequently studied in the economics of law enforcement literature, it is still only one of the multiple objectives that the criminal justice system seeks to further. Thus, the analysis here should be interpreted as providing a starting point for reviewing our implicit assumptions about how we socially rank various types of discrimination. Studying the impact of discrimination on other enforcement goals, e.g. administrative cost minimization and harm prevention (Friehe and Tabbach (2013) and Mungan (2017)), may reveal additional insights.

7 Appendix

In the proofs below, a superscript of $*$ refers to equilibrium values.

Proof of Proposition 1. (i) That there is a single equilibrium implies that $Q_W(x) > Q_W(x^*_W) = q_W$ for all $x > x^*_W$. Thus, $Q_B(x) \geq Q_W(x) > q_W \geq q_B$ for all $x > x^*_W$. Moreover,

$$Q_B(x^*_W) \geq Q_W(x^*_W) = q_W \geq q_B \tag{19}$$

where the first inequality holds strictly when G_B FOSD G_W, and the second inequality holds strictly when enforcers have racist preferences. Therefore, $Q_B(x) > q_B$ for all $x \geq x^*_W$ whenever there is discrimination, which implies that $x^*_B < x^*_W$, since $Q_B(x^*_B) = q_B$.

(ii) Examples demonstrating these possibilities are depicted in figure 2.■

Proof of Proposition 2. Discrimination implies that $x^*_B \neq x^*_W$, and thus $x^*_R \neq x^m$ for some $R \in \{B, W\}$. Thus, per lemma 1, $\theta^*_R > \theta^m_R$ for some $R \in \{B, W\}$, and, $\theta^*_R \geq \theta^m_R$ for $R \in \{B, W\}$ by definition (i.e. since θ^m_R is the minimum crime rate). Therefore $(1-\phi)\theta^*_W + \phi\theta^*_B > \Theta^m$.■

Proof of Proposition 3. In what follows the equilibrium threshold rules are expressed as $x^*_R(\lambda_R, \mu_R)$ to capture the fact that they respond to the liability regime.

First, note that, any targeted suspicion threshold (including $Q_R(x^m)$) for race R, denoted $\overline{q}_R$, can be achieved through, and can only be achieved through,

liability regimes that satisfy:

$$\frac{\psi_R + \lambda_R}{\psi_R + \lambda_R + \pi_R + \mu_R} = \overline{q}_R; \text{ or}$$
$$\mu_R = \frac{1-\overline{q}_R}{\overline{q}_R}\lambda_R + \kappa_R \text{ where } \kappa_R \equiv \frac{1-\overline{q}_R}{\overline{q}_R}\psi_R - \pi_R \tag{20}$$

Next, note that if $G_B \equiv G_W$, it follows that $x_B^*(\lambda_B, \mu_B) = x_W^*(\lambda_W, \mu_W)$ can only be obtained if the same threshold suspicion is targeted, i.e. $\overline{q}_R = \overline{q}$ for both $R \in \{B, W\}$ for some $\overline{q} \in (0,1)$. Note also that when $G_B \equiv G_W$, the presence of discrimination implies that either $\psi_B < \psi_W$, $\pi_B > \pi_W$, or both. Thus, whenever $\overline{q}_B = \overline{q}_W$, it follows that $\kappa_B < \kappa_W$, which, in turn, implies via (20) that

$$\mu_B - \frac{1-\overline{q}}{\overline{q}}\lambda_B < \mu_W - \frac{1-\overline{q}}{\overline{q}}\lambda_W \tag{21}$$

Manipulating this inequality yields:

$$0 < (1-\overline{q})[\lambda_B - \lambda_W] + \overline{q}[\mu_W - \mu_B] \tag{22}$$

which requires that either $\mu_B < \mu_W$, $\lambda_B > \lambda_W$, or both.

On the other hand, when G_B FOSD G_W, it follows that $x_B^*(\lambda_B, \mu_B) = x_W^*(\lambda_W, \mu_W)$ only if $q_B(\lambda_B, \mu_B) = \overline{q}_B > q_W(\lambda_W, \mu_W) = \overline{q}_W$ for some pair $\overline{q}_B, \overline{q}_W \in (0,1)$. This similarly implies that $\kappa_B < \kappa_W$, since $\overline{q}_B > \overline{q}_W$. This reveals, via (20), that

$$\mu_B - \frac{1-\overline{q}_B}{\overline{q}_B}\lambda_B < \mu_W - \frac{1-\overline{q}_W}{\overline{q}_W}\lambda_W; \text{ or} \tag{23}$$
$$\frac{\overline{q}_B - \overline{q}_W}{\overline{q}_W}\lambda_W < (1-\overline{q}_B)(\lambda_B - \lambda_W) + \overline{q}_B(\mu_W - \mu_B)$$

Thus, it follows that either $\mu_B < \mu_W$, $\lambda_B > \lambda_W$, or both.

These observations together reveal that eliminating double standards requires setting $\mu_B < \mu_W$, $\lambda_B > \lambda_W$, or both.Thus, race-neutral liability regimes cannot eliminate discrimination.■

Proof of Proposition 4. First, note that when the liability regime is race-neutral, it follows, via proposition 1-(i), that $x_W^*(\lambda, \mu) > x_B^*(\lambda, \mu)$. Next, express the equilibrium total crime rate as: $\Theta(\lambda, \mu) = \phi\theta_B^*(x_B^*(\lambda, \mu)) + (1 - \phi)\theta_W^*(x_W^*(\lambda, \mu))$. Because $\frac{\partial x_R^*}{\partial \lambda} > 0 > \frac{\partial x_R^*}{\partial \mu}$ for $R \in \{B, W\}$, and because deterrence is single peaked and maximized at x^m, it follows that $\frac{\partial \Theta^*(\lambda,\mu)}{\partial \lambda} < 0$ whenever λ and μ are such that $x_B^*(\lambda, \mu) < x_W^*(\lambda, \mu) \leq x^m$, and $\frac{\partial \Theta^*(\lambda,\mu)}{\partial \mu} < 0$ whenever λ and μ are such that $x^m \leq x_B^*(\lambda, \mu) < x_W^*(\lambda, \mu)$. Thus, no liability regime can be crime minimizing unless they generate threshold rules such that

$$x_B^*(\lambda, \mu) < x^m < x_W^*(\lambda, \mu) \tag{24}$$

Part (i): Note that when there exists no taste-based discrimination, $q_B = q_W$ for all race-neutral liability regimes, and all liability regimes that satisfy (20) for some $\overline{q} = q_B(\lambda, \mu) = q_W(\lambda, \mu)$ achieve the same level of crime. Thus, the crime rate can be expressed as a function of the targeted level of threshold $\overline{q}$, i.e. $\Theta^*(\lambda, \mu) = \widehat{\Theta}(\overline{q}(\lambda, \mu))$. Moreover, the previous observations that $\frac{\partial \Theta^*(\lambda,\mu)}{\partial \lambda} < 0$ when $x^*_W(\lambda, \mu) = x^m$ and that $\frac{\partial \Theta(\lambda,\mu)}{\partial \mu} > 0$ when $x^*_B(\lambda, \mu) = x^m$ together imply that $\frac{d\widehat{\Theta}(Q_W(x^m))}{d\overline{q}} < 0 < \frac{d\widehat{\Theta}(Q_B(x^m))}{d\overline{q}}$, and therefore there exists $q^P \in (Q_W(x^m), Q_B(x^m))$ which minimizes crime among all threshold suspicions. This, in turn, implies that crime is minimized by all combinations of λ and μ which generate this threshold suspicion.

Part (ii): Suppose a liability regime (λ, μ) generates threshold rules which satisfy (24) (otherwise, the liability regime cannot be crime-minimizing, as previously noted). For any such (λ, μ) pair there exists $\delta > 0$ such that the pair $(\lambda', \mu') \equiv (\lambda + \delta, \mu + \delta \frac{1-q_W(\lambda,\mu)}{q_W(\lambda,\mu)})$ results in less crime than (λ, μ), because $x^*_W(\lambda, \mu) = x^*_W(\lambda', \mu') > x^m > x^*_B(\lambda', \mu') > x^*_B(\lambda, \mu)$ where the second inequality holds because δ is chosen to be sufficiently small. This, in turn, implies that $\theta^*_B(\lambda', \mu') < \theta^*_B(\lambda, \mu)$ and $\theta^*_W(\lambda', \mu') = \theta^*_W(\lambda, \mu)$, and therefore $\Theta^*(\lambda', \mu') < \Theta^*(\lambda, \mu)$. Thus, no finite λ, μ pair minimizes crime.

To see that the increases in the crime rate due to taste-based discrimination can be completely eliminated in the limit, consider liability regimes of the form: $(\lambda, \widehat{\mu}(\lambda))$ where $\widehat{\mu}(\lambda) = \lambda \frac{1-q^P}{q^P}$ and q^P denotes the total crime minimizing q when there is no taste-based discrimination (as defined in part (i)). It follows that $\lim_{\lambda\to\infty} q_W(\lambda, \widehat{\mu}(\lambda)) = \lim_{\lambda\to\infty} q_B(\lambda, \widehat{\mu}(\lambda)) = q^P$, thus, $\lim_{\lambda\to\infty} \theta^*_{R\in\{B,W\}}(\lambda, \widehat{\mu}(\lambda)) = \theta^P_{R\in\{B,W\}}$.■

Proof of Lemma 2. Discrimination against blacks implies that the applicable thresholds denoted x^d_R, are such that $x^d_W > x^d_B$. When there is non-discriminatory enforcement, as previously noted, any $x^n \in (\underline{x}, \overline{x})$ can be obtained as the threshold rule for both races through an appropriate liability regime. Thus, because deterrence is single-peaked and maximized by the threshold rule x^m_R for $R \in \{B, W\}$, it follows that if $x^m_W \geq x^d_B$, switching to the non-discriminatory regime where $x^n = x^m_W$ leads to lower crime, and if $x^m_B \leq x^d_W$, switching to the non-discriminatory regime where $x^n = x^m_B$ leads to lower crime. These follow, because in both cases, the crime rate for one race is reduced without increasing the crime rate for the other race. If neither inequality holds, it follows that $x^m_W < x^d_B < x^d_W < x^m_B$, in which case, switching to a non-discriminatory regime where $x^n \in (x^d_B, x^d_W)$ leads to less crime, since the crime rate among both races is reduced.■

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